Nishi Kanta Paul

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Portfolio

Research Interests

Medical AI & Imaging · Multimodal AI / VLMs · Explainable AI · Computer Vision · NLP for Healthcare · Medical Image Segmentation

Education

B.Sc. in Computer Science and Engineering *2020 – 2024*
Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh
CGPA: **3.749 / 4.00** • Ranked **7th out of 100**

Research Experience

Voluntary Researcher *Feb 2025 – Present*
Univ. of Alabama at Birmingham & Temple University (Remote, USA)
Project: AI-Driven Parkinson's Disease Screening via Voice Analysis

- Designing a novel deep learning framework incorporating temporal feature modeling and adaptive feature fusion for voice-based PD detection
- Applying contrastive learning to improve discrimination between healthy and PD-affected speech; *first-author manuscript in preparation*

Research Intern *Dec 2025 – Feb 2026*
Elite Research Lab
Project: MedVQA-X — Medical Multimodal AI

- Contributed to a multimodal Medical Visual Question Answering (MedVQA) system integrating medical images and clinical text for clinical decision support
- Assisted in dataset preprocessing, annotation, and experimentation with vision-language models (VLMs)

Undergraduate Thesis Researcher *2023 – 2024*
AUST & Monash University (Fellow Institute), Dhaka, Bangladesh
Project: Video-Based Driver Distraction Detection — Bangladesh Perspective

- Collaborated with a Monash University PhD researcher; applied transfer learning across CNN architectures with dataset curation and cross-architecture benchmarking
- Co-authored a peer-reviewed paper published at **IEEE ICACCESS 2024**

Publications

[C1] Md. M. Kabir, Md. A. Nawar, M. I. Remon, **N. K. Paul**, H. Bhuiyan, M. M. Hoque, "Video-based Driver Distraction Detection using Transfer Learning: Bangladesh Perspective," *IEEE iCACCESS*, 2024. IEEE Xplore

Under Review & Manuscripts in Preparation

- **N. K. Paul** et al., "Comparative Analysis of Hybrid Deep CNN Models in Pulmonary Disease Classification," *under review* — ICEFronT: IEEE Int. Conf. on Engineering and Frontier Technologies 2026
- **N. K. Paul** et al., "Parkinson's Disease Screening via Voice Analysis," *first-author, in preparation*

- **N. K. Paul** et al., "StatAdapt-Loss: Adaptive Loss for Imbalanced Medical Image Segmentation via Skewness-Kurtosis Estimation," *first-author, in preparation*
- **N. K. Paul** et al., "MedExplain: ShinovaAI — Explainable Deep Learning for Breast Cancer Diagnosis from Mammography," *first-author, in preparation*

Scholarships & Academic Honors

Scholarship for Outstanding Academic Performance *Fall 2020*
Awarded for CGPA **3.98** and ranking **1st** in batch, AUST.

Khondakar Ahmed Muktadir & Mosammat Khodeza Begum Scholarship *Fall 2022*
Merit-based scholarship for sustained academic excellence, AUST.

Technical Skills

Languages	Python, JavaScript, TypeScript, C++
AI / ML	PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers
Web	React, Next.js, FastAPI, Django, Flask
Databases	PostgreSQL, MySQL, MongoDB
Tools	Git, Linux, Docker, LaTeX, Postman

Teaching Experience

Adjunct Lecturer, Dept. of CSE — *International Standard University, Dhaka* *Jan 2026 – Present*
Teaching Robotics, IoT, and Competitive Programming; designed lab exercises and course materials.

Competitive Programming Mentor (Remote) — *Shohoj Coding* *Sep 2024 – Dec 2025*
Mentored 50+ students in algorithms and data structures; designed structured curriculum for skill progression.

Industry Experience

Software Engineer I — *Brain Station-23 PLC, Dhaka* *Mar 2025 – Present*
Developed full-stack solutions for identity verification (OneID) and vehicle tracking (ObhaiVTS) using React, FastAPI, Django, and PostgreSQL.

Competitive Programming

ICPC Asia Dhaka Regional Contest 2023 — Rank: **116th**, Team: AUST_Tactless
Intra-AUST Programming Contest, Spring 2022 — Rank: **1st**, Team: AUST_Durnibar

Profiles: Codeforces · LeetCode · Toph · CodeChef

References

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